Medical Hypotheses 109 (2017) 119–125

Contents lists available at ScienceDirect

Medical Hypotheses

journal homepage: www.elsevier.com/locate/mehy

M. biceps femoris – A wolf in sheep's clothing: The downside of a lower limb injury prevention training

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ABSTRACT

Both, hamstring and ACL injuries are among the most typical injuries, particularly in change of direction and high speed running sports. They're also difficult to treat. Therefore, in the past few years, sports medicine practitioners and exercise scientists have mainly been focusing on the development and implementation of preventive programs in order to reduce the number of lower limb injuries, mainly by improving knee alignment. A number of studies have been able to prove the success of these training interventions, which are mainly addressing sensorimotor abilities and plyometric activities. The number of non-contact hamstring injuries has nevertheless been on the rise, particularly in sports like soccer and football. Therefore, the purpose of the following article is to introduce the hypothesis that the above-mentioned training interventions have a massive influence on the activation patterns on the targeted muscle group, and on the *M. biceps femoris* in particular. Muscle function and the resulting internal load are directly related to muscle architecture at the insertion. Training induced adaptations in hamstring activation patterns can thus lead to an increased injury susceptibility. In this case, a simulation model that directly relates to an acute deceleration maneuver provides valuable insights into the function of the biceps femoris muscle, especially when the rate of activity and the muscle geometry at the insertion area are taken into consideration. We conclude that there needs to be a greater individualization of prevention programs, especially in regards to anatomical requirements, in order to further reduce injury rates in elite sports. Moreover, it would also seem reasonable to apply a similar approach to aspects of chronic pain such as chronic non-specific low back pain.

Introduction

Muscle or tendon injuries are among the most common non-contact injuries in sports [1]. In a long-term study, Ekstrand and co-workers revealed that there were eight injuries per 1000 h of player exposure, which adds up to about 50 injuries per team per season [1]. Up to 90% of all injuries affect the lower extremities [2]. Injuries to the hamstring muscles (HAMS) are among the most frequent with a prevalence of up to 25%, depending on the type of sport [2,3]. Today it is commonly accepted that injuries of this muscle group most notably occur in sports with high speed running, frequent changes of direction, and intermittent sprinting and deceleration phases [4]. Hamstring injuries are generally associated with long time-loss and a lengthy rehabilitation process [5,6]. The economic loss incurred by a top-level club for an injured player being out for one month is estimated at €500.000 [7]. This corresponds to approximately €1.5 million for an injury that affords an 80 day time-loss [6,7]. In about 83% of hamstring injuries *M. biceps femoris* is affected, with 11% occurring to the *M. semimembranosus* and only 5% to the *M. semitendinosus* [8]. Around 16% of all hamstring injuries were re-injuries, all of which affected the *M. biceps femoris* [8]. By comparing muscle function via MRI, Schuermans et al. [9] could demonstrate that eccentric strength endurance was significantly lower in subjects with a history of hamstring injuries. In addition, a symmetrical recruitment pattern between *M. biceps femoris* and *M. semitendinosus* indicates a reduced activation economy in the hamstring muscle group. These findings implicate that the susceptibility for hamstring injuries is directly related to the complexity and the synergistic activation of these two muscles, in which case *M. biceps femoris* needs to compensate for the lack of endurance capacity of *M. semitendinosus* [9].

Looking at the anatomy of the *M. biceps femoris* muscle, it is commonly known that its distal tendon inserts at the proximal head of the *Fibula*. Its proximal end is two headed, the long head originates at the *ischial tuberosity*, the short head at the *linea aspera* of the *Femur*. Tubbs et al. [10] have identified a medial and a lateral part at the insertion, each with an anterior and a posterior component. They also found

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http://dx.doi.org/10.1016/j.mehy.2017.10.002
Received 30 March 2017; Accepted 5 October 2017
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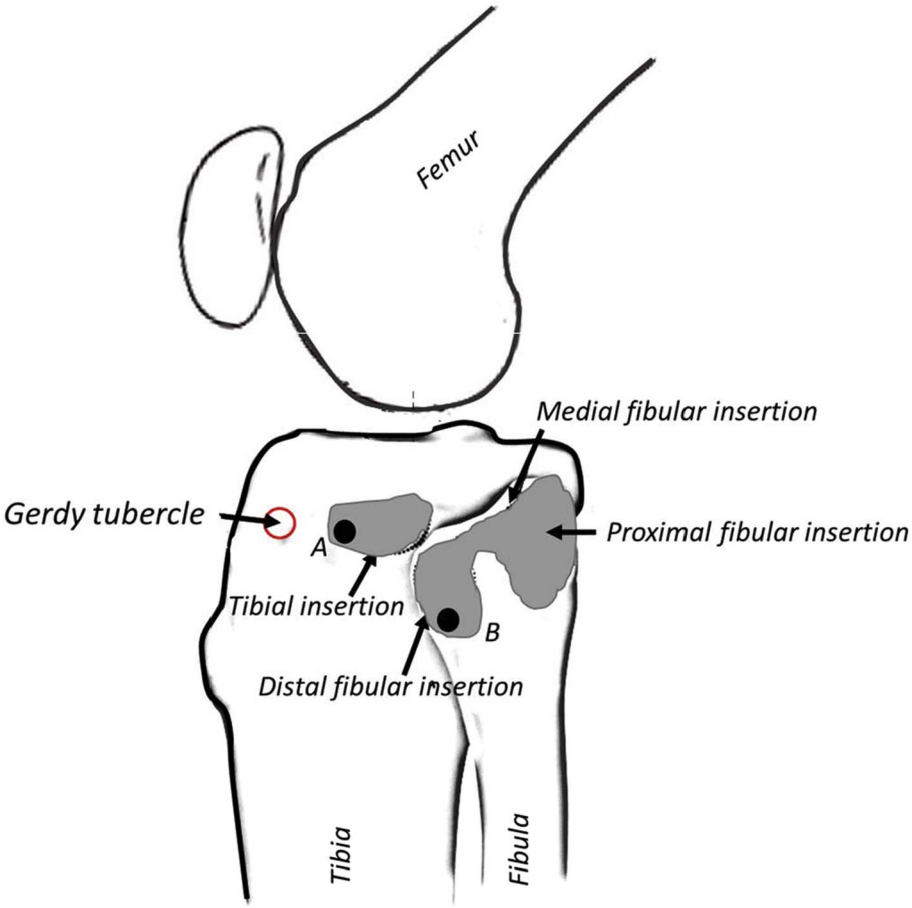

Fig. 1. Simplified lateral view showing the insertion areas of the *M. biceps femoris* according to Branch and Anz [11]. A and B exemplary insertional architecture for the chosen model.

additional insertions into the *M. popliteus* tendon as well as the arcuate popliteal ligament. They therefore assume that there's a possible synergy between the *M. biceps femoris* and the *M. popliteus* that may be crucial for knee function and joint stability [10]. Recent findings suggest that the insertion of the *M. biceps femoris* may be even more complex than previously assumed. Apart from the proximal and distal fibular attachments there is may be an additional medial attachment at the *Fibula*, as well as a tibial insertion [11]. These respective insertions add up to a reasonably sized area (see Fig. 1).

The actual knee joint function is primarily determined by the musculoskeletal geometry, e.g. the spatial arrangement of the muscle at the respective segments and the joints crossed. The *M. biceps femoris*, as mentioned above, is a bi-articular muscle. Its function is fundamental when it comes to transmitting force and velocity to more distal body segments [12–14]. But likewise, not only are bi-articular muscles able to decrease the angle between two body segments, but under certain conditions also to increase it. This means that a bi-articular muscle with different starting conditions and equal activation can lead to entirely contrastive system behavior. Referring to the example of the leg it signifies that the contraction of such a muscle on the one hand leads to an extension of the leg or the knee and on the other hand to a shortening of the leg or a flexion of the knee. The *M. biceps femoris* can actually change from being a knee flexor to becoming a knee extensor [15–20]. This phenomenon is commonly known as Lombard's paradox [17]. A closed kinematic chain is necessary to implement this behavior. This means that the foot has to be fixed on the ground as well as the hip over an abutment. However, studies on activation patterns of the affected muscles have revealed that mechanical conditions alone are not sufficient to induce such a reverse function. Comparative studies on jumping have demonstrated that in spite of identical mechanical conditions, the *M. biceps femoris* activation pattern turned out to be far from similar within the examined cohorts [20]. In subjects with a long jump training experience, the primary activation pattern could be maintained throughout the complete range of motion, e.g. from landing in a deep squat position to take-off while the foot was planted.

In lesser trained subjects the activation only took place in the very last moment before take-off and the knee almost fully extended. In his study of abrupt stopping maneuvers after high speed running, Vallée proved that the activity of the *M. biceps femoris* in elite athletes strongly depends on their respective sport [21]. Gruber and colleagues [22] confirmed an increase of the *M. biceps femoris* activity after several weeks of sensory motor training, however, without a significant change in kinematic activity in the lower extremity. These findings are supported by Nagano et al. [23] who found that after five weeks of plyometric training, the activity of *M. biceps femoris* immediately before landing was significantly higher than before the intervention. Interestingly, English and colleagues discovered in animal studies on cats an overabundance of nerve endings in feline muscle compared with the number of motor units, particularly in the *M. biceps femoris* [24]. They too confirmed that *M. biceps femoris* acts as a synergist in knee extension, albeit with a considerable inter subject variability from which he concludes that there must be an influence of some type of sensory feedback. The nerve endings to motor units ratio indicates an enhancement in the sensory system. These findings would underline that English's assumptions also apply to the human body and that knee extension induced by the *M. biceps femoris* does not solely depend on mechanical requirements, but especially on a sensory motor feedback (see Fig. 2).

The interconnection with *M. popliteus* appears to be yet another unexplained issue when it comes to mechanical behavior of the *M. biceps femoris*. Coupling mechanisms that at least partially inhibit the activation of *M. biceps femoris* are being discussed in this context

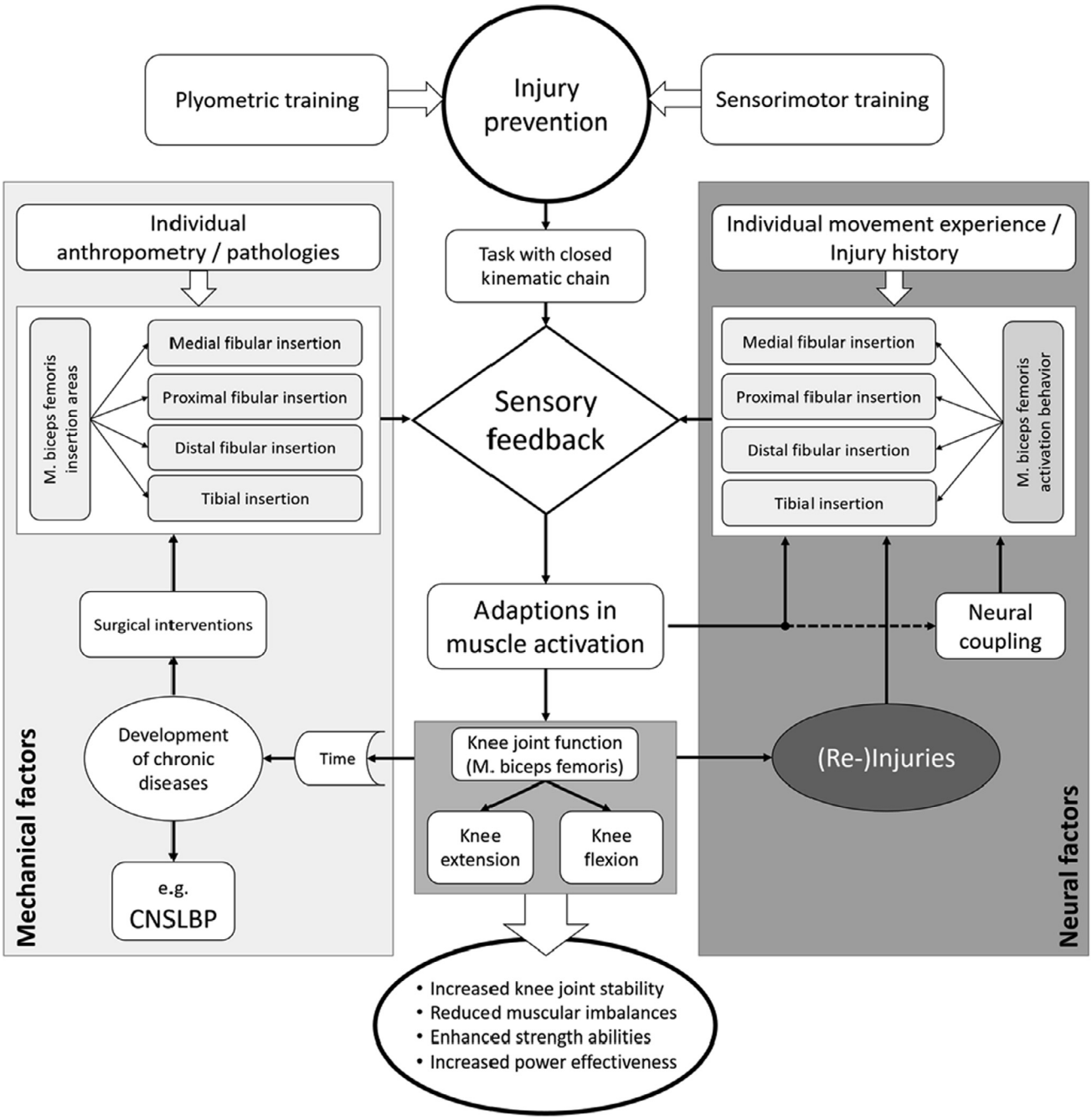

Fig. 2. Scheme of activation during tasks with closed kinematic chain. Consequences of *M. biceps femoris* activity regarding an increased and/or prematurely activation pattern in dependency of selected determinants (CNSLBP: Chronic non-specific low back pain).

[10,20]. The architecture at the insertion of the *M. biceps femoris* as detected in the studies of Tubbs et al. as well as Branch and Anz [10,11] also appears to raise further questions concerning mechanical behavior that have not yet been addressed to this day. The latest outcomes suggest that mechanical as well as anthropometric characteristics are highly responsible for function and behavior of *M. biceps femoris*. This also applies to neural mechanisms. All of these factors substantially affect the internal load of the muscle. The cited literature brings attention to the high injury susceptibility particularly in *M. biceps femoris*. According to long term investigations, highly valid amounts of information has been be gathered from the field of sports [1,3,8,9,25]. In addition to that, current results show that despite a high focus on prevention programs, the hamstring injury rate has risen by 4% in recent years [26].

The hypothesis

There are numerous sport specific programs for injury prevention that are targeting the stabilization of the knee. This particular focus is mainly due to the high prevalence of ligamental injuries in the knee joint [27]. Measures with a high proportion of sensorimotor and plyometric exercises, that have been developed since 2005, are particularly efficient when it comes to ACL injuries [28–30]. These types of interventions have also been applied in the form of eccentric exercises for the prevention of hamstring injuries. Furthermore, it could also be demonstrated that a variation of exercises can lead to a change in muscle length and hypertrophy, in addition to refined activation patterns [31,32].

There is evidence that an eccentric and isolated strength training as

well as complex hamstring exercises can reduce the incidence of hamstring injuries [33,34]. However, while various injury prevention programs have been implemented in the last decade, we have still not managed to reduce the incidence of hamstring injuries. Under the given circumstances we assume that the internal muscle load is not only substantially affected by the geometry at the insertion, but also through changes in the activation patterns after a specific training intervention. The geometry at the insertion directly influences muscle leverage and thus may enable phenomena such as Lombard's paradox in the knee joint. This means that a change in activation patterns can induce concentric but also eccentric muscle activity in the existence of abutments of closed kinematic chains. An eccentric muscle behavior, during which the muscle is mechanically lengthened despite its contraction, directly increases the passive muscle load. Consequently, the risk of injury is also increased. We further assume that the geometry at the insertion of the *M. biceps femoris* is subject to individual peculiarities or pathologies, while previous injuries also lead to changes in the distribution of force within the area of insertion.

Evaluation of the hypothesis

The underlying hypothesis deals with a high-risk muscle behavior that may cause foreseeable harm to potential participants. The most elegant way to circumvent this problem is to demonstrate the relevant behavior in a computer simulation. By simulating a critical maneuver that is commonly present in many sports, our aim is to demonstrate the exemplary influence of both the muscular insertion and the level of neural muscle activation on the muscular loading pattern. The movement pattern we selected constitutes the single leg and partially unanticipated stopping maneuver after running. In order to simulate this activity we created a model of the neuro-musculo-skeletal system for human locomotion that relates to realistic segmental and muscular features and illustrates basic prerequisites as they appear in such stopping maneuvers (see Fig. 3). This simplified muscle-driven forward dynamic simulation model consists of six essential leg muscles but only the *M. biceps femoris* underwent changes to both, its insertion and the level of activation. The actual simulation refers to segmental muscle behavior during an one legged stopping maneuver in an 80 kg male subject after running at 2 m/s. Joint configuration at the beginning of the stopping movement was calculated after video analysis of football players with the hip angle being set at 40°, the knee angle at 30° and the dorsiflexion of the foot also at 40°. The configuration of the anthropometric model was conducted based on "calcman" and Hill-Type muscles according to Günther [35]. The specific architecture at the insertions of *M. rectus femoris*, *M. vastus intermedius*, *M. tibialis anterior*, *M. soleus* and *M. gastrocnemius* were adapted from Horsman et al. [11,36]. Joint specificity was adapted from Lohmann [37]. The setting up of the model and the simulation approach is based on the work of Andrassy [38]. The selected running speed leading to the actual stop rests on the findings of Ertelt and Vallée as well as Hase and Stein [21,39,40]. The model as such, along with the simulation was conducted with MatLab/SimMechanics (Mathworks R, Version 2012).

Segmental kinematics during the simulated stopping maneuver turned out to be highly dependent on the respective point of insertion of the *M. biceps femoris* (see Fig. 4). When this muscle is activated at 100%, the model shows a noticeable knee extension during the stopping maneuver, given the area of insertion is located at the tibia (see Configuration A in Fig. 1 and Fig. 4A). Reversely, with the insertion located in the distal fibular area (see Configuration B in Fig. 1 and Fig. 4B), a noticeable knee flexion occurred within the same time frame.

The influence of the *M. biceps femoris* on the dynamics of knee extension was assessed by varying the level of neural activation in the muscle. We thus assigned three characteristic rates of activity at 0%, 50%, and 100%. Considering the insertion configuration A (see Fig. 1 and Fig. 4A) and the level of neural activation of *M. biceps femoris*, we were able to demonstrate the role of the muscle in knee extension (Fig. 5).

In case of 0% of *M. biceps femoris* activation, knee extension was induced by the "traditional" extensors, namely *M. rectus femoris* and *M. vastus intermedius*, whereas at 100% of *M. biceps femoris* neural activation the actual knee extension took place in a much greater volume and a considerably shorter time.

Since all this information relies on assertions based on simplified models, it is necessary for these findings to be verified in experimental

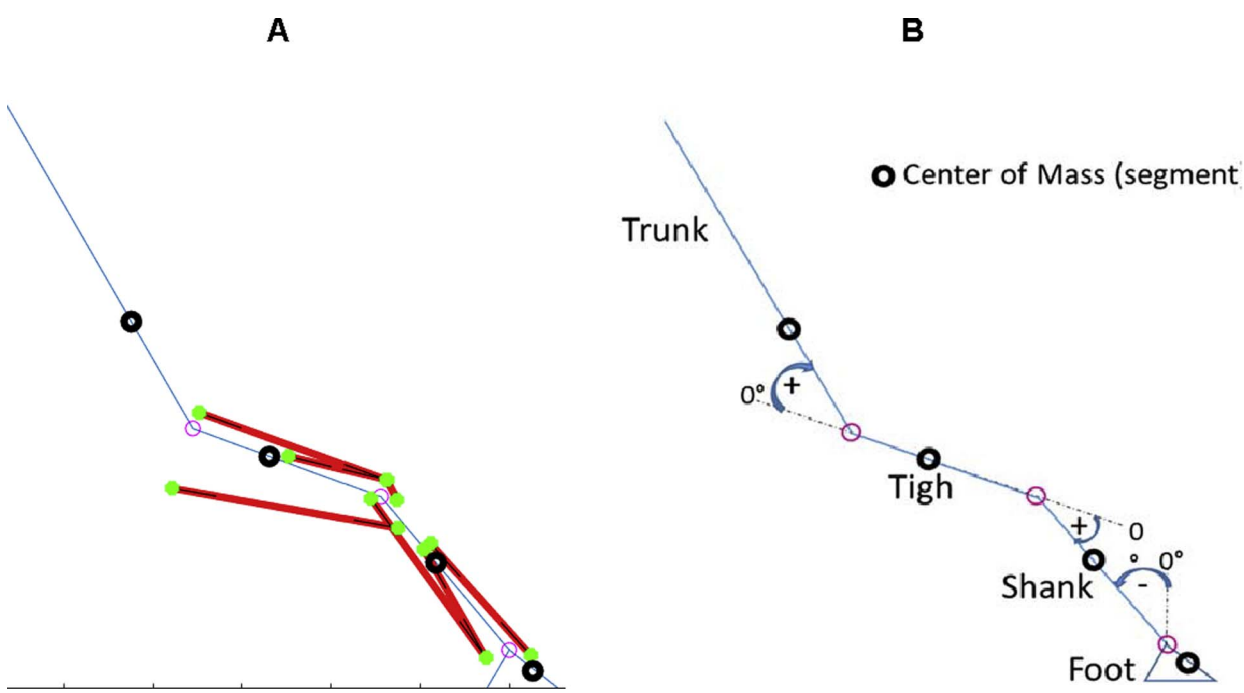

Fig. 3. A: Anthropometric muscle configuration based on "calcman" and Hill-Type muscles according to Günther [35]. Bodyweight: 80kg. Insertional geometry of the *M. rectus femoris*, *M. vastus intermedius* , *M. tibialis anterior*, *M. soleus*, and *M. gastrocnemius* according to Horsman et al. [36]. *M. biceps femoris* calculated and adapted by Branch and Anz [11]. Dynamics of movement stop according to Lohmann [37]; Model set-up by Andrassy [38]. Situation at the moment of impact: hip angle 40°; knee angle 30°; dorsiflexion −40°. B: angular definitions.

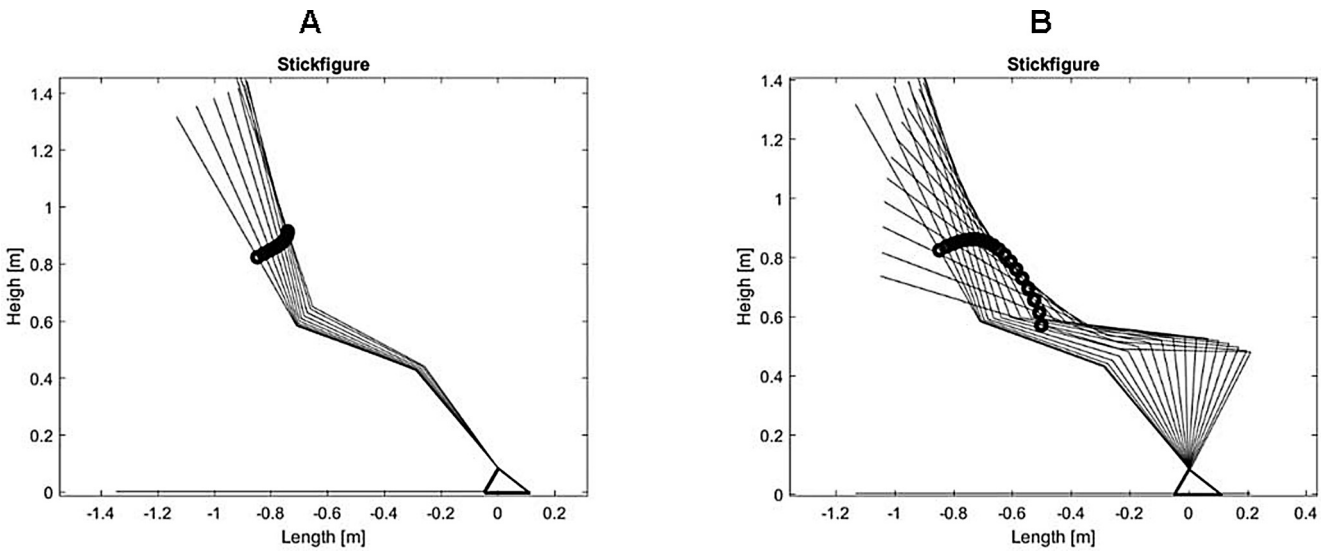

Fig. 4. Behavior of a lower extremity model during a simulated one legged stopping maneuver after running and depending on the insertional geometry of the *M. biceps femoris* A/B. Simulation time 400 ms starting on first contact. Starting parameters at impact: hip angle 40°; knee angle 30°; dorsiflexion −40°. Speed ModelKSP v0x = 2m/s; v0y = −0.3m/s.

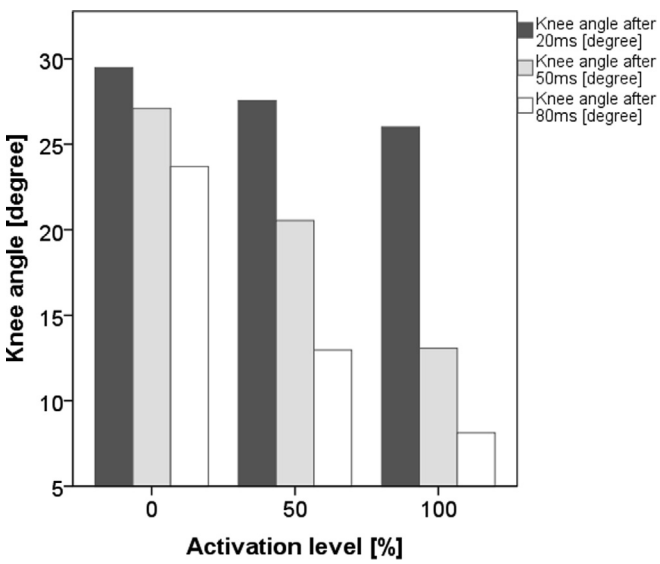

Fig. 5. Knee angle depending on rate of activation of the *M. biceps femoris* at 20ms, 50ms and 80ms after initial ground contact.

studies. This means that stopping maneuvers need to be conducted in a controllable setting. Therefore, sledge jumps appear to be an adequate option due to the fact that they allow pre-setting of joint angles and limiting of degrees of freedom [20,41,42]. These examinations would then have to be complemented by EMG and MRI analyses in order to acquire relevant data concerning rates of activity, but also anthropometry, especially in terms of insertion geometry. In addition to that, sport related skill level and a potential history of injury also need to be taken into account. Vallée and Ertelt [20,40] point out that the correlation between the actual sport and the muscular activity in the *M. biceps femoris* is highly significant.

Relevance for clinical practice

Despite an increased implementation of special interventions to decrease the risk of lower limb muscle injuries, the number of hamstring injuries has been on the rise. Current evidence based training interventions on injury prevention in the lower extremities are mainly focused on sensorimotor and plyometric exercises. Our simulation shows that a change in neural activation patterns as well as differences in the insertion area lead to changes in the knee joint function of *M. biceps femoris*. If this holds true in an actual trial, then the induced innervation changes by injury prevention programs will significantly increase the internal muscle load and impose an increased injury risk. This, in turn, leads to the assumption that by diagnosing a "specific" geometry of insertion (e.g. MRI), along with an individual pattern of activation behavior in sport specific movement tasks (EMG), requires an individually adapted training intervention. Such an intervention would then be capable to actually reduce the individual risk of injury. Our hypothesis seems particularly relevant in the vast field of knee arthroplasty due to a massive impact on the geometry of the insertion of *M. biceps femoris*. Depending on the damage to the knee joint, the type of knee implant and the actual way the surgery is performed, part of the tibial plateau needs to be removed. It is advised that the *Fibula* be left out in this procedure. However, practical experience shows that in some cases during joint reconstruction the *Fibula* is damaged at least in its proximal part. In both of these cases there will be an impact on the areas of insertion, as demonstrated by Branch and Anz [11]. Last but not least, the outlined mechanism may be greatly relevant regarding the symptoms of chronic non-specific low back pain. Numerous studies have been able to prove the immanent role of the *M. biceps femoris* [19,43,44]. The activation pattern in connection with the geometry of insertion could lead to new insights, especially in diagnostics and therapy of those patients classified as dominantly nociceptive [45,46].

Discussion

Our simple, multibody muscle model was primarily constructed in order to answer questions concerning the high incidence of injuries of the *M. biceps femoris*. Despite its simplicity, results of the simulation were able to show basic connections between muscle activation function, and geometry of insertion that are all directly related to internal muscle load. The level of muscle loading can be regarded as an indicator for the degree of injury risk. The extended insights into the complex insertional geometry of the *M. biceps femoris* by Tubbs et al. [10] as well as by Branch and Anz [11] point to a very complex way of muscle functioning. Depending on the respective points of insertion, mechanical muscular leverage changes in an identical movement task. Changes in leverage at the insertion area inevitably lead to changes in the corresponding lever arms of a bi-articular muscle. Lombard [17], Wiemann [15] and Ertelt et al. [18,47] have pointed out the critical leverage within a muscle of this type. Once its leverage reaches a

particular magnitude, it will inevitably lead to knee extension which could be showed convincingly in Fig. 4A. A change of geometry induces a dramatically changed behavior of the leg (see Fig. 4B) and is able to completely reverse movement dynamics. The insertion areas of the *M. biceps femoris*, as pointed out by Branch and Anz, [11] allow a completely different interpretation of the results presented by English [24]. Despite the fact that these findings have been gathered from feline samples, they are now able to generally explain the insertional issues in this muscle. The results presented by English refer to an enormous sensorimotor feedback system in the muscle that appears to be overly extensive compared to other leg muscles. According to English, neuromuscular feedback of this muscle is twice as great compared to other leg muscles. However, this apparent overabundance of sensory feedback makes a lot of sense when the different insertional options are taken into account. As demonstrated in our model, different insertions lead to different joint behaviors. In this context, such a large area of insertion definitely requires some vast and intricate sensing. According to Tubbs [10] there also appear to be coupling mechanisms with other muscles, e.g. the *M. popliteus*, that can partially inhibit or at least influence the *M. biceps femoris* activation. Taking mechanical features into account, a partial inhibition of muscular activity would surely have an influence on internal load. Moreover, Winter and Yack [48] have been able to prove significant variations in inter-subject activation patterns in human gait studies. This would then demonstrate that reply to respective tasks cannot be considered uniform but relies on the geometry of insertion. This, however, depends on individual anthropometric measures [11].

Gruber and colleagues [22], as well as Nagano et al. [23] were able to prove an increased activity in the *M. biceps femoris* following a specific plyometric or sensorimotor exercise protocol. Furthermore, Vallée [21,49] and Ertelt [20] report a differently structured, however uniform activation pattern depending on movement literacy as well as exercise experience. Along with Gruber [22] they also point out that lower extremity movement kinematics rarely change despite activity alterations of the *M. biceps femoris*. This, in turn, implicates that force transmission to the knee doesn't necessarily increase following an exercise induced, enhanced activation of the *M. biceps femoris* which would then entail a change in kinematics. It rather affects other areas of insertion that may previously not have been engaged in this manner. An activation increase that coincides with force distribution to various areas of insertion will to some extent result in a hampering of their respective function. Force that's being fed via the distal tibial insertion tends to initiate an extension of the knee due to its proximity to the center of rotation as described by Lombard [17]. On the other hand, the knee will most likely be flexed when the force transmission takes place via the fibular insertion areas. Both functions would thus cancel each other out which would instead only lead to an increase in muscle load. The rate of activation also needs to be seen in this context. With regard to the respective geometry at the insertion, even a minor muscle activation may prove sufficient to decisively contribute to knee extension, as has been shown in Fig. 5. The large lever ratio between origin and insertion of the *M. biceps femoris* is primarily responsible for an effect on knee extension at a low rate of activation, as has been presented in Fig. 4A. Case A (see Fig. 4A) shows an origin to insertion ratio of approx. 4:1, while in case B (see Fig. 4B) the ratio is only about 2:1. This means that even a minor increase of activity in those muscle fibres that introduce force via the tibial insertion area will lead to a deviant functional behavior in the knee joint and a significantly increased muscle load.

Current prevention programs in sports as well as in general prevention strongly focus on sensorimotor and plyometric interventions. This is especially true in those sports that show a high risk for knee injuries and where these types of special exercise programs are being widely applied to help with knee stabilization. Stabilization of the knee leads to a decreased ACL injury risk. Evidence concerning the effectiveness of these interventions has been widely documented. On the other hand, especially in recent years the susceptibility for hamstring injuries, particularly in professional sports, has been significantly on the rise [3,26,35]. In numerous cases, this phenomenon has been put down to the changing on-field demands. However, the model study we have presented in this paper suggests a different context. This directly refers to the geometry of insertion, as well as the activation of those muscle fibers whose force transmission mechanisms may lead to deviant joint behaviors. The high re-injury rate of the *M. biceps femoris* can also be viewed in this context. Scarring could lead to a compromised activation of individual muscle components, resulting in alterations of force transmission mechanisms and consequently changes in knee joint function. Force transmission that is being relocated to tibial insertion areas would induce an increased knee extension. In stopping maneuvers, this leads to a high eccentric load which is always accompanied by a high injury risk.

Therefore, the key for the adaptation of the activation behavior lies in a specific training [20,22]. If the phase displacement of the activation pattern of the M. *biceps femoris* can be corrected to the “normal pattern”, the described negative effects could then be prevented. We suppose that this therapeutic approach may have a major benefit for the muscular resilience in the affected athletes and economics in sport specific settings.

Conflicts of interest

This paper represents our own opinion which is not affected by any third party.

Acknowledgements

We would like to thank the German Research Foundation (DFG) and its experts for their broad vision and the assessment of this topic. Despite numerous applications from us within the past 10 years, this topic has always been declared as irrelevant and our originality and know-how have been denied. The project was not supported by the DFG.

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